DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 8: The Efficient Frontier

The Feasible Set · Min. Variance Portfolio · The Markowitz Bullet

August 10, 2026

Today's Agenda

1. From one portfolio to the feasible set

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2. The minimum variance portfolio

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4. More assets can only help

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4. More assets can only help
5. The frontier as a quadratic program

Learning Objectives

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- Define the efficient frontier; explain why the lower branch is never worth holding.
- Explain why adding more assets can only improve the frontier.
- Write the frontier as a constrained quadratic optimization problem.

Every Recipe You Could Cook

Ingredients to Dishes

N assets are like N ingredients; a portfolio is a recipe – a proportion of each.

Every Recipe You Could Cook

Ingredients to Dishes

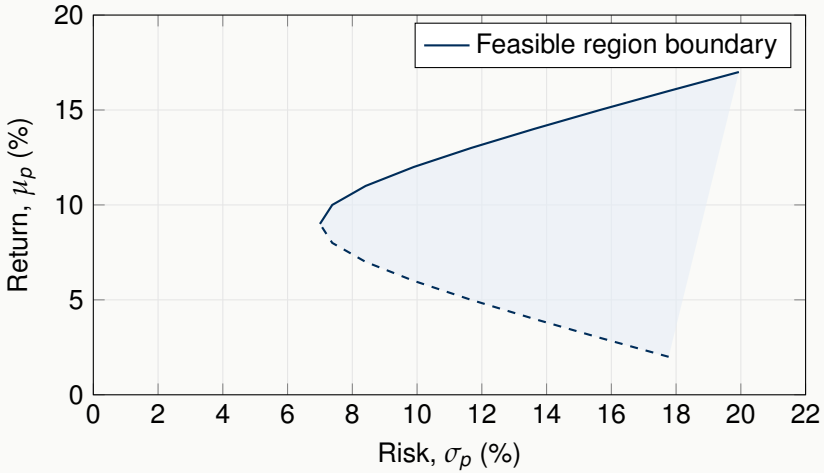
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Every Recipe You Could Cook

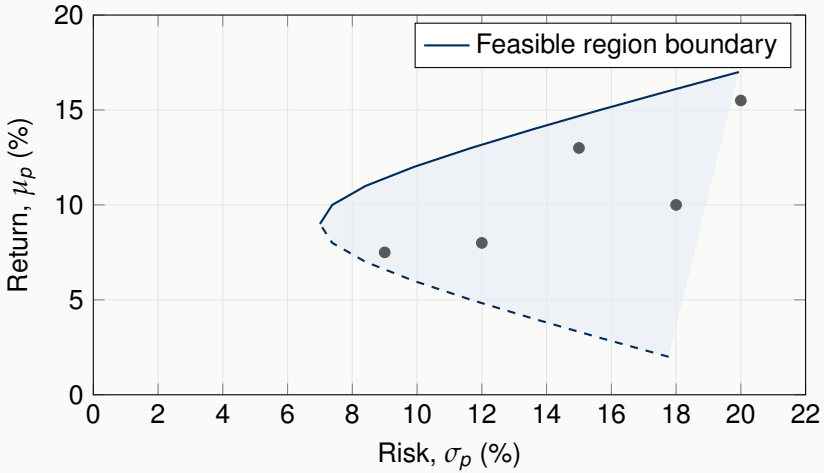
Ingredients to Dishes

N assets are like N ingredients; a portfolio is a recipe – a proportion of each. Varying proportions traces out every achievable risk-return combination. The feasible set is exactly this: everything you can "cook up" from your assets.

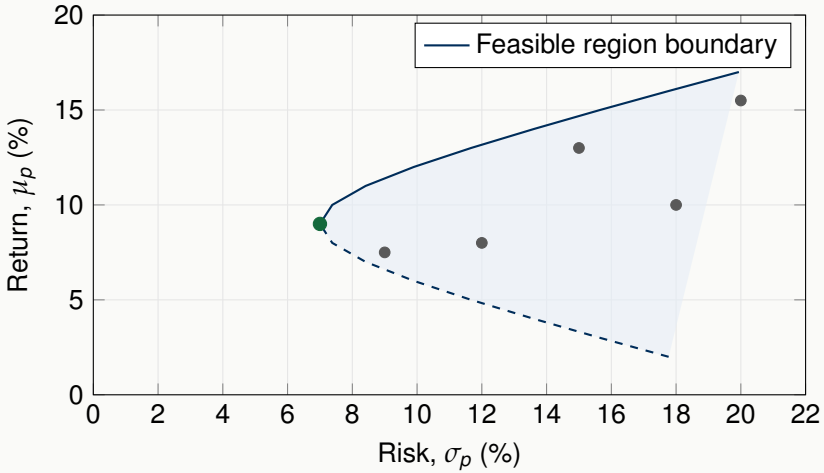
Visualizing the Feasible Set



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The Minimum Variance Portfolio

Definition

The point in the feasible set with the lowest σ_p, regardless of return:

$$\min_{w} \; w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1$$

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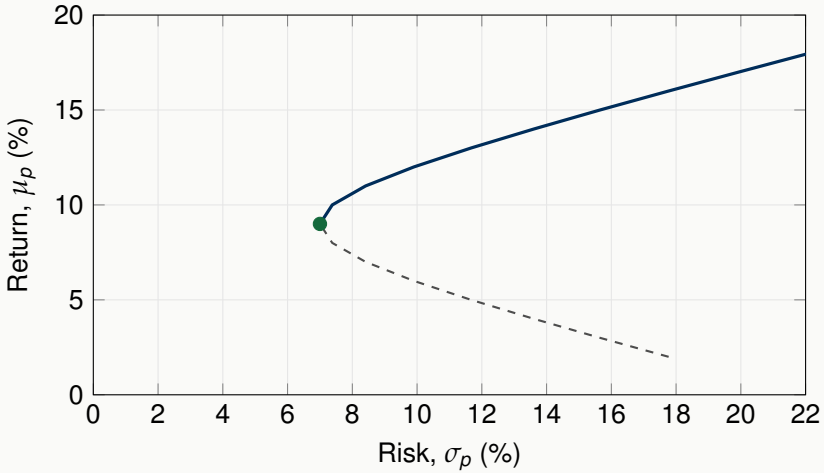
Needs *no* information about μ – only Σ. Covariances are typically far more stable to estimate than expected returns.

Efficient Frontier: Definition

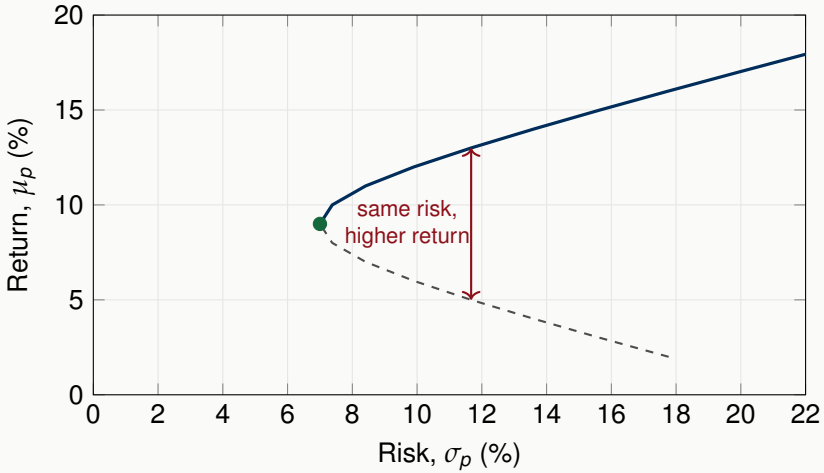
Definition

The **upper** boundary of the feasible set, from the MVP upward: highest return for each risk level (equivalently, lowest risk for each return level).

Efficient vs. Inefficient



Efficient vs. Inefficient



Why the Lower Branch Is Worthless

Dominated, Not Just Different

Two portfolios, same risk ($\approx$ 11.67%): upper branch gives 13%, lower gives only 5%.

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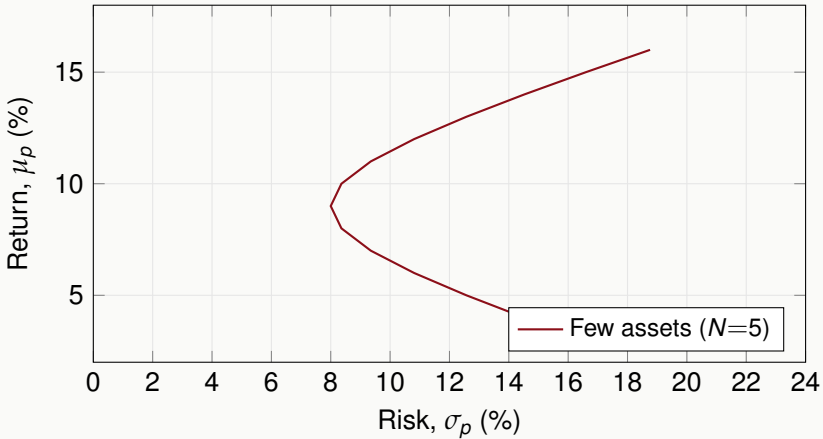
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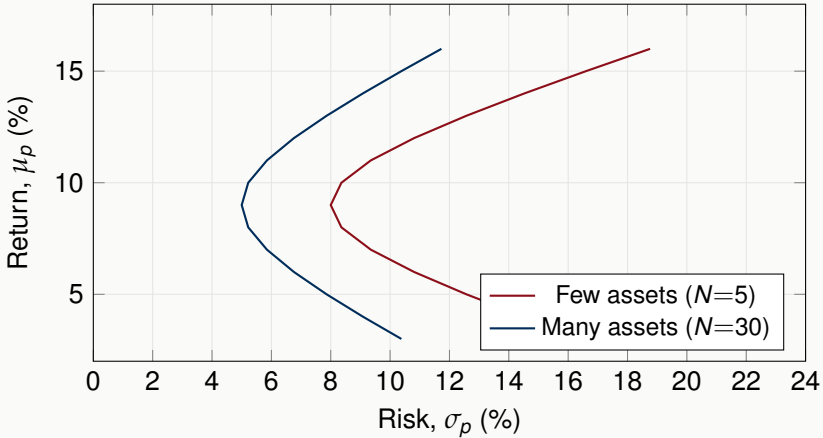
Dominated, Not Just Different

Two portfolios, same risk ($\approx 11.67\%$): upper branch gives 13%, lower gives only 5%. Every risk-averse (or even risk-neutral) investor prefers more return for the same risk. The lower branch is **dominated** – never economically relevant, even though it's mathematically feasible.

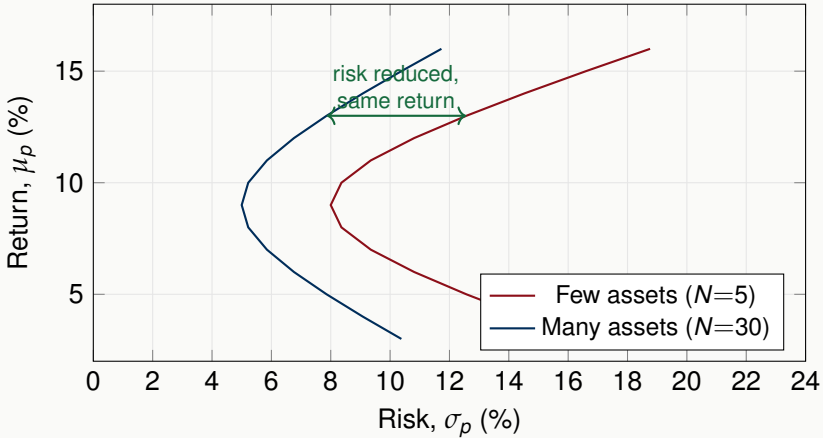
More Assets Can Only Help



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Adding Assets Never Hurts

Why

Every portfolio achievable with N assets stays achievable with $N+1$ (give the new one zero weight).

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Every portfolio achievable with N assets stays achievable with $N+1$ (give the new one zero weight). So the feasible set can only **grow**. More assets $\Rightarrow$ frontier moves left (or stays put) – never right.

The Markowitz Problem

The Frontier as an Optimization Problem

$$\min_{w} \; w^\top \Sigma w \quad \text{s.t.} \quad \mu^\top w = \mu^*, \quad \mathbf{1}^\top w = 1$$

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Quadratic objective, linear constraints.

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Quadratic objective, linear constraints. No simple closed form once you add $w_i \geq 0$ – solved numerically. Exactly what you'll do in this week's Lab, on real NIFTY50/S&P500 data.

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- The **MVP** is the leftmost point – needs only Σ, not μ.
- The **efficient frontier** is the upper boundary; the lower branch is always dominated.
- More assets can only push the frontier left – never worsen it.
- The frontier solves a **quadratic program**.

Check Your Understanding

Discuss (2 minutes)

1. Why does an individual asset almost always lie *inside*, not on, the feasible region?

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3. Why does the MVP not need μ, while other frontier points do?

Looking Ahead

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Two-Fund Separation – Lecture 9